

DroneArjuna (DA)

As-Built Implementation Status Report

Generated directly from the current codebase — reflects what is implemented, not what is planned

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Cross-referenced against: DA-Project-Timeline-AI-Adjusted (Phase 1–6 task register)

1. Executive Summary

This report documents what is **actually implemented** in the DroneArjuna system as of this date, derived from direct inspection of the backend (FastAPI/Python) and frontend (React/TypeScript) applications, the database migration history, the automated test suite, and the deployment configuration. It is organized against the same six-phase structure used in the AI-Adjusted Project Timeline so status can be read phase-by-phase.

Area	Status	Notes
Phase 0 — Scaffold & Infra	Complete	Full container stack, 9 services, all networked together
Phase 1 — Auth + MAVLink core	Complete	JWT auth, RBAC, MAVLink connection manager, telemetry pipeline, live streaming
Phase 2 — Drone Master + Mission Planner	Complete	Full CRUD for types/instances/payloads/vessels/config templates; mission CRUD, validation, survey-grid, fleet assignment
Phase 3 — Advanced Flight + Visualisation	Partial	2D map, geofence enforcement + auto-RTL, deconfliction/fleet-assign solver present; no 3D visualisation found
Phase 4 — Drone Inventory	Partial	Catalogue/compare/search/threat-systems endpoints exist; wraps Drone Master data — full-text search engine is deployed and indexed, but the full knowledge-base workflow is marked as a later-phase deliverable
Phase 5 — Drone Analyst (AI/CV)	Partial (scaffold only)	Job registry, telemetry stats/series, model registry, detection-result storage all present; no object-detection/inference/video-processing code found
Phase 6 — Hardening/Security/UX	Partial	Password policy, token blacklist, rate limiter, audit log, input sanitiser all implemented; CI pipeline / accessibility pass / load testing not verified

2. Backend — Implemented Modules

2.1 Application Entrypoint

The FastAPI application exposes a Swagger/API documentation page and mounts routers for authentication, drone control, drone master data, inventory, flight planning, and analyst functions, plus a root health-check endpoint.

On startup the application: connects to the primary database with automatic retry, initializes the message queue, starts a geofence-breach auto-return-to-launch consumer, starts the telemetry recorder (which creates the time-series storage tables), starts a listener for vessel position feeds, seeds a default administrator account if none exists, bulk-indexes inventory data into the search engine, and starts a background auto-connect task.

2.2 Drone Control Module

Endpoints exposed by this module:

Method	Function	Purpose
GET	List ports	Enumerate available serial/USB/UDP connection ports
POST	Auto-connect	Trigger auto-connect scan across ports
GET	Status	Connection status of all drones
POST	Connect	Open MAVLink connection to a drone
POST	Disconnect	Close a drone connection
GET / POST	Geofence	Get / set geofence for a drone
POST	Command	Send flight command (arm/disarm/RTL/etc.)
GET	Telemetry snapshot	Current telemetry snapshot
GET	Telemetry gauges	Dashboard gauge values
GET	Telemetry history	Historical telemetry for replay
POST / DELETE / GET	Simulation controls	Start / stop / query a simulated flight
Live stream	Telemetry stream	Live telemetry push connection per drone

Key capabilities:

- **MAVLink connection manager** — async per-drone connection management (UDP/TCP/serial/HF), one task per drone
- **Telemetry processor** — parses incoming MAVLink messages by type, updates state, emits geofence breach/recovery events
- **Command controller** — validates commands against drone state, encodes/sends with retry and acknowledgement tracking
- **State manager** — in-memory hot cache bridging the connection layer and live-stream broadcaster; wraps proximity monitoring

- **Health monitor** — threshold-based alerting (battery, signal strength, GPS satellites, CPU) with auto-return-to-launch on critical battery
- **Telemetry recorder** — persists telemetry to the database (current-state snapshot plus an append-only history log)
- **Auto-connector** — polls for hardware appearance, retries disconnected drones
- **HF link adapter** — high-frequency radio datalink tier for ship-to-shore operations (priority queueing, degraded-link handling)
- **MAVLink broadcaster** — encodes real MAVLink v2 packets from simulated state, broadcasts for external ground-control-station interoperability
- **Mission simulator** — synthetic telemetry state machine (idle → armed → takeoff → flying → return/landing → landed)
- **Proximity monitor** — inter-drone separation checks (250m minimum), stale-telemetry exclusion
- **Vessel position feed** — ingests vessel GPS via the HF link, updates naval vessel records for dynamic home-basing

2.3 Drone Master Module

Full CRUD across five entity groups:

- **Drone types:** list/create/stats/get/update/delete
- **Drone instances:** list/create/get/update/status patch/delete/stale-cleanup/payload-assign/type-spec lookup
- **Vessels:** list/create/get/update/position-update/assign-drone/unassign-drone/delete
- **Payload types:** list/create/get/update/delete
- **Config templates:** list/create/get/update/delete/apply-to-drone

2.4 Drone Inventory Module

Provides: drone list, drone detail, quick-reference card, side-by-side comparison, full-text search, payload catalogue, and threat-system CRUD (read/write/notes gated by separate role permissions). Implemented as a first-version wrapper over Drone Master data with formatted display structures; a dedicated search engine is deployed and used for inventory indexing at startup.

2.5 Drone Flight Module

Provides: mission CRUD (list/create/get/update/status-patch/delete), mission summary (distance/time/battery estimate), mission validation (including airspace/geofence checks), mission upload to vehicle, live-sync to a connected drone, pre-flight simulation preview generation, multi-drone fleet assignment, and survey-grid generation from a polygon.

Key capabilities:

- **Mission planner** — validates missions against drone performance limits, uploads/executes missions, generates survey grids
- **Airspace service** — hardcoded India restricted-airspace data (restricted and caution zones per national aviation authority guidance) used for validation guards

- **Geo service** — geodesy utilities (great-circle distance calculations), battery-estimate constants, builds mission summaries
- **Fleet assignment solver** — multi-drone target assignment solver; default exact constraint-programming solver, with an optional experimental quantum-computing solver

2.6 Drone Analyst Module

Provides: analysis job create/list/detail/cancel, detection-result listing, model registry list/detail, and mission telemetry stats/time-series endpoints.

This module is a **first-version scaffold**: job registry, telemetry analytics, and model registry are implemented and backed by the database. No object-detection inference, video-frame processing, or background-worker infrastructure for machine-learning jobs was found — these remain later-phase deliverables per the project timeline.

2.7 Core — Auth, Security, RBAC

- **Authentication** — JWT-based auth using an OAuth2 password flow. Provides token issuance, current-user lookup, forced password setup, admin-only user registration/listing/password-reset (emails a temporary password), public forgot-password/reset-password flow (30-minute token expiry), and a full access-request workflow (public signup request; admin accept/reject/resend-email). A default administrator account is auto-provisioned if none exists.
- **Role-based access control** — Five roles: administrator, mission commander, flight controller, intelligence analyst, and viewer. Enforcement is hierarchical for the operational roles, with intelligence analyst treated as a non-hierarchical specialist role. Applied selectively to sensitive endpoints (e.g. threat-system write operations).
- **Security utilities** — password policy (length/complexity/blocklist), a cache-backed token blacklist for revocation, a cache-backed rate limiter (sliding window; login and command-rate presets defined), an append-only audit logger, and an input sanitiser (control-character stripping, HTML escaping, injection-pattern logging).

2.8 Data Models

Entity group	Key entities
Drones	Drone type (spec catalogue), drone instance (fleet record), drone configuration template
Missions	Mission, waypoint (with action types, altitude reference)
Users	User account, access request
Telemetry	Telemetry frame (current state), telemetry history (replay log), telemetry gauge, system metric — stored in the time-series database
Vessels	Naval vessel (position, sea state, HF modem configuration)
Payloads	Payload type (sensor and combat payload fields)
Threats	Threat system (UAV/radar/SAM/electronic-warfare knowledge base)

2.9 Live Streaming

One live telemetry stream connection is implemented, providing a continuous push of telemetry data per drone.

2.10 Database Migrations

15 schema migrations, from the initial schema (users/drone types/instances/missions/waypoints) through naval-vessel support, audit logging, payload tables (with one superseding revision replacing an earlier payload table design), access requests, soft-delete flags, password reset tokens, and a threat-systems table. Two parallel migration branches were later merged.

3. Frontend — Implemented Components

3.1 Application Shell

- **App shell** — top-level layout: navigation sidebar, top bar, notification drawer, camera window, error-boundary-wrapped workspace panels
- **Login screen** — login form plus a request-access signup flow
- **Password setup screen** — forced password-change screen
- **Sidebar** — navigation between workspaces: Fleet, Plan, Fly, Monitor, Settings
- **Top bar** — user info, notifications, connection/telemetry status, timezone display

3.2 Workspaces

Workspace	Contents
Fleet	Roster management, per-drone status card (battery/link/GPS/speed/vessel/payload), connect dialog with port scan
Fly	Live flight view, live map (markers plus mission overlay), instrument display (with geofence/regulatory-zone awareness), command panel, manual control panel, simulation launch and progress overlay
Monitor	Gauge dashboard, telemetry charts, system event log
Plan	Mission-editing map with geofence overlays, waypoint editor, live operations panel, multi-drone fleet assignment dialog
Settings	Drone type manager, drone instance manager, payload manager, vessel manager, user manager (including access-request approval)

Also present: shared components (camera window, confirmation dialogs, error boundaries, notification drawer, status badges, system log viewer), flight instrument widgets (altitude gauge, artificial horizon, battery gauge, speed gauge), and geofence/regulatory-zone/route-collision utilities.

No 3D visualisation capability exists yet — the advanced-flight-phase deliverables for 3D terrain, live 3D tracking, and a telemetry replay player are not implemented.

3.3 State Management

Client-side state stores cover: authentication, fleet/connection state, missions and waypoints, live per-drone telemetry (with automatic reconnect), naval vessels, connection health (latency/packet metrics across channels), notifications, a categorized system event log, access-request workflow state, and display timezone.

3.4 API Client

A shared HTTP client plus dedicated modules for authentication, drone control, flight/mission operations, drone master data, payloads, and vessels.

4. Testing

All automated tests are backend tests — **no frontend test files exist.**

- **~854 test functions across 42 test files**
- Coverage areas: authentication/RBAC/access-requests, MAVLink connection internals (command controller, auto-connector, broadcaster, health monitor, telemetry processor/endpoints, telemetry recorder, proximity monitor, HF link adapter/feed, vessel position feed), mission/geofence/deconfliction logic, drone-master CRUD, inventory and threat-systems, analyst functions, and infrastructure/search/system-health checks

5. Infrastructure

Service	Role
Primary database	Relational database with spatial (GIS) support
Time-series database	Telemetry hypertables (separate instance)
Database admin UI	Web UI for both database instances
Cache / in-memory store	Token blacklist, rate limiter
Message queue	Event bus (geofence-breach → auto-RTL consumer, etc.), with management UI
Object storage	Deployed for imagery/video storage; not yet used in code
Search engine	Inventory full-text search/indexing
Dev mail catcher	Local inbox for approval/reset emails during development
Backend application	API server; exposes MAVLink and vessel-position network ports; runs with elevated hardware access for USB passthrough
Frontend application	Web application

All services share a single private network with named persistent storage volumes.

6. Gap Summary vs. Project Timeline

Timeline Item	Status
3D visualisation scaffold, terrain draping, 3D tracking, replay player	Not implemented
Geofence enforcement + auto-RTL, multi-drone deconfliction	Implemented (event-driven consumer + proximity monitor + fleet solver)
Full search-engine inventory integration	Search engine deployed and indexed; full knowledge-base-style workflow marked as a later-phase deliverable
Threat system / order-of-battle knowledge base	Model + CRUD endpoints implemented
Background workers, object-detection, change-detection, video extraction	Not implemented
Job persistence, object-storage upload, analyst UI, telemetry dashboard	Job/telemetry backend present; object storage deployed but unused in code; frontend analyst workspace not confirmed
≥80% automated test coverage pass	~854 tests exist; coverage percentage not verified
Token blacklist + rate limiting	Implemented (cache-backed)
Accessibility pass, load testing, CI pipeline	Not found

This report reflects a point-in-time inspection of the system and should be re-generated as Phases 3–6 progress. It is derived entirely from code inspection, not from design documents or planning artifacts.